CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 3: Shares and Dividend

Topic-wise short notes with solved examples

A share is a small part of the capital of a company. This chapter covers the two values of a share, the dividend it earns and the return an investor actually gets.

1 Shares and their values

- The capital of a company is divided into small equal units called **shares**. A person who buys shares is a **shareholder**.
- **Nominal value (N.V.)**, also called face value or par value, is the value printed on the share. It does not change.
- **Market value (M.V.)** is the price at which the share is bought or sold in the market. It changes with time.

Share is	Meaning	Example for a ₹100 share
At par	M.V. $=$ N.V.	Sold at ₹100
At a premium (above par)	M.V. $>$ N.V.	Premium of ₹20: sold at ₹120
At a discount (below par)	M.V. $<$ N.V.	Discount of ₹15: sold at ₹85

- A share bought "at ₹20 premium" costs N.V. $+20$. A share bought "at 10% discount" costs N.V. -10% of N.V.

2 Dividend

- The part of the profit that a company distributes to its shareholders is the **dividend**.
- Dividend is always expressed as a percentage of the **nominal value**, never of the market value.

$$\text{Dividend on one share} = \frac{\text{rate of dividend} \times \text{N.V.}}{100} \qquad \text{Total dividend} = \text{number of shares} \times \text{dividend on one share}$$

- The money a person spends on buying shares is the **investment** (sum invested).

$$\text{Number of shares} = \frac{\text{sum invested}}{\text{M.V. of one share}}$$

Example 2.1 A man buys 400 shares of nominal value ₹25 each at a market value of ₹36 each. The company pays a dividend of 12%. Find his investment, his annual dividend and his percentage return.

SOLUTION

Investment $= 400 \times 36 =$ ₹14,400

Dividend on one share $= 12\%$ of ₹25 $=$ ₹3, so annual dividend $= 400 \times 3 =$ ₹1,200

Return $= \dfrac{1200}{14400} \times 100 = 8\frac{1}{3}\%$

Investment ₹14,400; dividend ₹1,200; return $8\frac{1}{3}\%$

Example 2.2 Find the number of ₹100 shares that can be bought for ₹8,800 when they are at a premium of ₹10, and the dividend received if the company pays 15%.

SOLUTION

M.V. of one share $= 100 + 10 =$ ₹110

Number of shares $= \dfrac{8800}{110} = 80$

Dividend $= 80 \times 15\%$ of ₹100 $= 80 \times 15$

80 shares; dividend ₹1,200

3 Return on investment

- The **return** (or yield) is the income as a percentage of the money actually invested.

$$\text{Return } \% = \frac{\text{dividend on one share}}{\text{M.V. of one share}} \times 100$$

- A very useful relation follows:

$$\text{Rate of dividend} \times \text{N.V.} = \text{Return } \% \times \text{M.V.}$$

Example 3.1 A company pays a dividend of 10% on its ₹50 shares. Find the return on the investment of a man who buys them at ₹62.50.

SOLUTION

Dividend on one share $= 10\%$ of ₹50 $=$ ₹5

Return $= \dfrac{5}{62.50} \times 100$

Return $= 8\%$

Example 3.2 Which is the better investment: 16% ₹100 shares at ₹80, or 20% ₹100 shares at ₹120?

SOLUTION

First: dividend ₹16 on an investment of ₹80; return $= \dfrac{16}{80} \times 100 = 20\%$

Second: dividend ₹20 on an investment of ₹120; return $= \dfrac{20}{120} \times 100 = 16\frac{2}{3}\%$

The first, 16% ₹100 shares at ₹80, is the better investment.

4 Selling and reinvesting

- Sale proceeds = number of shares sold × M.V. at the time of sale.
- If the proceeds are reinvested, find the new number of shares, then the new dividend, and compare with the old one.

Example 4.1 A man sells 60 shares (N.V. ₹100, paying 10%) at ₹120 each and invests the proceeds in ₹50 shares at ₹40 that pay 12%. Find the change in his annual income.

SOLUTION

Original income $= 60 \times 10 =$ ₹600

Sale proceeds $= 60 \times 120 =$ ₹7,200

New shares $= \frac{7200}{40} = 180$; new income $= 180 \times 12\%$ of ₹50 $= 180 \times 6 =$ ₹1,080

His annual income increases by ₹480.

5 Common mistakes

- Calculating dividend on the market value. Dividend is on the nominal value.
- Finding the number of shares by dividing by the nominal value. Divide the investment by the **market** value.
- Mixing up "rate of dividend" (on N.V.) with "return" (on M.V.).
- Reading "premium of 10%" as ₹10 for every share. It is 10% of the nominal value.

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